UNIT 2 · TOPIC 2.3

# Protecting Physical Spaces

Quiz

*Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.*

**Name:** **Date:** **Class:**

**1. Which of the following is an example of a managerial control used in physical security?**

- ○ A. A biometric fingerprint scanner installed on the server room door
- ○ B. A reinforced steel door with a deadbolt lock at the building entrance
- ○ C. A written policy restricting server room access to IT personnel only
- ○ D. A surveillance camera mounted above the lobby entrance

**2. A key card system logs every entry into a restricted research lab, recording the badge ID, timestamp, and door accessed. This system is best classified as which type of security control?**

- ○ A. Managerial
- ○ B. Technical
- ○ C. Operational
- ○ D. Physical

**3. Which of the following best describes the primary security function of a mantrap (security vestibule)?**

- ○ A. To detect unauthorized access attempts through embedded motion sensors and trigger an alarm
- ○ B. To prevent tailgating by permitting only one person to pass through at a time before the next door opens
- ○ C. To create an auditable log of every individual who enters a restricted zone
- ○ D. To alert security personnel when a door is held open beyond a set time threshold

**4. You are on the physical security team at Xtensr Research Labs. After assessing the newly acquired facility, you recommend installing biometric palm scanners at the server room entrance. Which statement best explains how this control mitigates the risk of unauthorized server room access?**

- ○ A. It deters attackers by making the facility appear heavily monitored and difficult to breach
- ○ B. It provides a recovery mechanism by recording who was present in the server room during a breach
- ○ C. It detects unauthorized entry attempts in real time and sends an alert to the security team
- ○ D. It prevents unauthorized entry by verifying identity using a physical attribute unique to each person, which cannot be shared or transferred

**5. Xtensr Labs installs CCTV cameras covering all entry points of the newly acquired research lab. Which of the following best explains how this control mitigates physical security risk?**

- ○ A. It deters potential attackers who know they are being recorded and provides forensic evidence to support investigation after an incident
- ○ B. It prevents unauthorized individuals from entering the facility by blocking physical access to entry points
- ○ C. It authenticates the identity of employees and grants or denies entry based on recognized faces
- ○ D. It encrypts communication between security checkpoints to prevent interception by adversaries

**6. Xtensr Labs requires all visitors to present a government-issued ID to the receptionist, sign a physical visitor log, and wear a temporary visitor badge for the duration of their visit. Which of the following best explains how this managerial control package mitigates physical security risk?**

- ○ A. It prevents all unauthorized persons from entering the facility by creating a physical barrier at reception
- ○ B. It automatically alerts the security team when a visitor's identity cannot be verified against an internal database
- ○ C. It establishes accountability by creating an auditable record of visitor identity and access, deterring malicious visitors and supporting investigation if an incident occurs
- ○ D. It encrypts visitor credentials to protect against identity theft during the check-in process

**7. During the physical vulnerability assessment of Xtensr Labs' new facility, an assessor observes that employees frequently hold the badge-access door open for colleagues immediately behind them rather than requiring each person to badge in individually. Which control most directly addresses this vulnerability, and why?**

- ○ A. Installing additional CCTV cameras at badge-access doors, because detection of tailgating incidents creates accountability
- ○ B. Requiring annual security awareness training on tailgating risks, because informed employees are the most cost-effective control against social engineering at doors
- ○ C. Updating the acceptable use policy to explicitly prohibit piggybacking, because managerial controls set the behavioral standard employees are expected to follow
- ○ D. Implementing a mantrap or security vestibule at high-security entrances, because it physically enforces single-person entry regardless of employee social behavior

**8. Xtensr Labs uses badge card readers at all perimeter doors but secures interior server rooms with standard combination locks. A security auditor flags this configuration as a control gap. Which evaluation of this configuration is most accurate and aligned with NIST physical access control principles?**

- ○ A. The configuration is sufficient because perimeter controls represent the highest-priority threat vector and the combination locks provide an adequate secondary barrier
- ○ B. The configuration introduces an accountability gap: combination locks produce no access log, so unauthorized or inappropriate server room access by insiders cannot be detected or investigated after the fact
- ○ C. Combination locks are more secure than badge readers for interior rooms because knowledge-based authentication (a memorized code) is harder to clone than a physical card
- ○ D. The configuration is compliant with NIST PE-3 because it uses two distinct control types across different access boundaries

**9. A budget-constrained research facility must select between two physical security proposals: (1) biometric scanners on all 12 interior doors, or (2) a formal managerial control package, a mandatory visitor escort policy, a contractor access procedure, and an access review schedule, enforced by the security manager. Which evaluation of these options is most consistent with NIST guidance on security control selection?**

- ○ A. Option 1 is always superior because technical controls are inherently stronger and more reliable than managerial controls
- ○ B. Option 1 eliminates insider threat risk entirely, making the managerial controls in Option 2 redundant
- ○ C. Option 2 may be cost-effective and rapidly deployable but requires reliable human enforcement and complementary technical controls to close the gaps that managerial controls alone cannot address
- ○ D. Option 2 is superior because NIST guidance establishes that managerial controls always take precedence over technical controls in the control selection hierarchy

**10. Which of the following best illustrates the concept of layered physical security (defense in depth)?**

- ○ A. Installing a single, high-grade biometric scanner at the main building entrance

- ○ B. Placing surveillance cameras at every entrance and corridor in the building
- ○ C. Using a perimeter fence, then a badge-access door, then a locked server cage in sequential order
- ○ D. Requiring employees to memorize long access codes for all interior doors

**11. In a layered physical security model, what is the term for the strategy of placing multiple sequential, independent access barriers so that an adversary must defeat each one to reach a target asset?**

- ○ A. Zero trust
- ○ B. Redundancy
- ○ C. Separation of duties
- ○ D. Defense in depth

**12. Which of the following is NOT typically considered a layer in a physical defense-in-depth security model?**

- ○ A. A perimeter fence with motion-activated security lighting
- ○ B. A network intrusion detection system monitoring server traffic for anomalies
- ○ C. A security guard stationed at the building reception desk
- ○ D. A badge-access door leading to the server room

**13. During the physical vulnerability assessment of Xtensr Labs' new facility, your team documents four vulnerabilities: no perimeter fencing, an unlocked and unmonitored loading dock, no visitor management system, and a server room secured only by a standard key lock. Which combination of layered controls would most effectively address the greatest number of these vulnerabilities?**

- ○ A. Install CCTV at the loading dock and replace the server room key lock with a badge reader
- ○ B. Install alarm systems throughout the facility and create a visitor sign-in policy
- ○ C. Post a security guard at the loading dock and issue key cards to all employees for the server room
- ○ D. Add a perimeter fence, install a badge reader at the loading dock, implement a visitor management system, and upgrade the server room to biometric access control

**14. Xtensr Labs is particularly concerned about insider threats, specifically, employees accessing server rooms they have no legitimate business reason to enter. Which layered set of controls would most effectively mitigate this risk?**

- ○ A. Install surveillance cameras in all hallways and post a security guard outside the server room door
- ○ B. Require all employees to sign a policy acknowledging that server room access is restricted to authorized personnel
- ○ C. Implement role-based badge access so only personnel with authorized roles can unlock server room doors, log all access attempts with timestamps, and conduct weekly access log reviews
- ○ D. Install physical locks on all server racks inside the room as the primary safeguard

**15. The newly acquired Xtensr Labs facility has a single main entrance with no access control and interior rooms secured only by standard pin-tumbler locks. Which assessment of this security posture and recommended layered control additions is correct?**

- ○ A. The posture is acceptable; pin-tumbler locks are a recognized industry standard sufficient for interior research rooms
- ○ B. The posture is insufficient at both boundaries; implementing a mantrap or badge reader at the main entrance and role-based badge access on interior rooms would add prevention at the perimeter and accountability inside
- ○ C. The highest risk is at the perimeter; installing a badge reader at the main entrance and adding a visitor sign-in system would adequately address all identified vulnerabilities

○ D. The highest risk is insider threat; surveillance cameras and a door-ajar alarm system would best close both the perimeter and interior gaps

**16. NIST SP 800-53 recommends applying physical security controls at multiple nested boundaries: site perimeter, building perimeter, work area, and asset level. Xtensr Labs has strong outer controls, a perimeter fence, a staffed guard post, and CCTV at all entry points, but no asset-level controls: server racks are unlocked and storage media is unencrypted. Which evaluation of this configuration is most accurate?**

○ A. The configuration is sufficient because perimeter controls represent the highest-impact threat vector and prevent the vast majority of attacks

○ B. The configuration aligns with NIST PE-3 because it uses three distinct outer-layer control types, satisfying the layered control requirement

○ C. Asset-level controls are unnecessary in a research setting where all staff have undergone background checks and are considered trusted

○ D. The configuration creates a single-layer failure mode: an adversary or insider who bypasses or is already inside the perimeter gains unrestricted access to all assets, with no inner control to limit the damage

**17. Xtensr Labs has a layered physical security architecture: badge readers at building and server room entries, CCTV at all entry points, motion sensors active after hours, and a visitor log at reception. Despite these controls, a recent audit found that former employees' badges were not deactivated within 24 hours of termination on three separate occasions. Which analysis most accurately identifies the critical control gap and the most appropriate corrective layer to add?**

○ A. The gap is in the managerial layer; a formal, policy-enforced offboarding procedure that mandates immediate badge deactivation upon termination, with verification by HR and the security team, is the missing control

○ B. The gap is in the detection layer; deploying additional cameras with facial recognition at entry points would identify and flag former employees attempting to enter

○ C. The gap is in the technical layer; replacing badge readers with biometric scanners would prevent former employees from accessing the building because their biometric profiles would be removed upon termination

○ D. The gap is in the physical layer; rekeying all locks after each employee termination would prevent former employees from returning

**18. Xtensr Labs must present a physical security investment recommendation to executive leadership. The choice is between: (1) a single high-investment solution, biometric scanners on all 12 interior doors, or (2) a layered architecture investing in a perimeter fence, badge readers at two main entries, a visitor management system, locked server racks, and a formal offboarding procedure. The facility's threat model includes both external adversaries and potential malicious insiders across 6 interior zones and 2 exterior entry points. Which argument best supports the layered architecture in alignment with NIST physical security principles?**

○ A. Layered controls are always less expensive per security zone and therefore represent better budget allocation

○ B. Biometric scanners at every interior door fully eliminate tailgating risk and constitute their own layered architecture because they require two factors: presence and biometric identity

○ C. A layered architecture ensures that the compromise of any single control, whether by tailgating, credential theft, or an insider, does not expose all assets; different control types address different threat actors and failure modes simultaneously across all boundaries

○ D. The layered architecture is superior because NIST requires organizations to implement controls from all three families, managerial, operational, and technical, and Option 1 uses only a technical control